

Substituting $[T]$ from Eq. (5.1.10) and $[k']$ from Eq. (5.1.8) into Eq. (3.4.22), ($[k] = [T]^T [k'] [T]$) we obtain the general transformed global stiffness matrix for a beam element that includes axial force, shear force, and bending moment effects as follows:

$$[k] = \frac{E}{L} \times \begin{bmatrix} AC^2 + \frac{12I}{L^2} S^2 & \left(A - \frac{12I}{L^2}\right) CS & -\frac{6I}{L} S & -\left(AC^2 + \frac{12I}{L^2} S^2\right) & -\left(A - \frac{12I}{L^2}\right) CS & -\frac{6I}{L} S \\ AS^2 + \frac{12I}{L^2} C^2 & \frac{6I}{L} C & -\left(A - \frac{12I}{L^2}\right) CS & -\left(AS^2 + \frac{12I}{L^2} C^2\right) & \frac{6I}{L} C & \\ 4I & \frac{6I}{L} S & -\frac{6I}{L} C & 2I & & \\ AC^2 + \frac{12I}{L^2} S^2 & \left(A - \frac{12I}{L^2}\right) CS & \frac{6I}{L} S & & & \\ AS^2 + \frac{12I}{L^2} C^2 & -\frac{6I}{L} C & \frac{6I}{L} S & & & \\ \text{Symmetry} & & & & & \end{bmatrix} \quad (5.1.11)$$

The analysis of a rigid plane frame can be undertaken by applying stiffness matrix Eq. (5.1.11). A **rigid plane frame** is defined here as a series of beam elements rigidly connected to each other; that is, the original angles made between elements at their joints remain unchanged after the deformation due to applied loads or applied displacements.

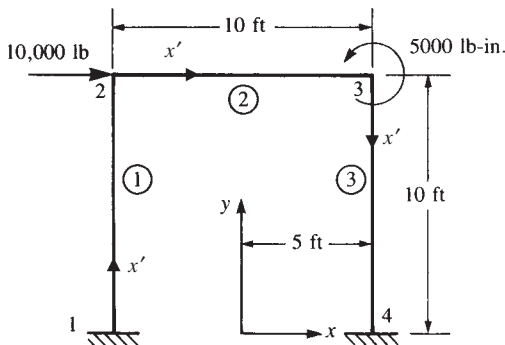
Furthermore, moments are transmitted from one element to another at the joints. Hence, moment continuity exists at the rigid joints. In addition, the element centroids, as well as the applied loads, lie in a common plane ($x-y$ plane). From Eq. (5.1.11), we observe that the element stiffnesses of a frame are functions of E , A , L , I , and the angle of orientation θ of the element with respect to the global-coordinate axes. It should be noted that computer programs often refer to the frame element as a beam element, with the understanding that the program is using the stiffness matrix in Eq. (5.1.11) for plane frame analysis.

5.2 Rigid Plane Frame Examples

To illustrate the use of the equations developed in Section 5.1, we will now perform complete solutions for the following rigid plane frames.

EXAMPLE 5.1

As the first example of rigid plane frame analysis, solve the simple “bent” shown in Figure 5–4.



■ Figure 5–4 Plane frame for analysis, also showing local x' axis for each element

SOLUTION:

The frame is fixed at nodes 1 and 4 and subjected to a positive horizontal force of 10,000 lb applied at node 2 and to a positive moment of 5000 lb-in. applied at node 3. The global-coordinate axes and the element lengths are shown in Figure 5-4.

Let $E = 30 \times 10^6$ psi and $A = 10 \text{ in.}^2$ for all elements, and let $I = 200 \text{ in.}^4$ for elements 1 and 3, and $I = 100 \text{ in.}^4$ for element 2.

Using Eq. (5.1.11), we obtain the global stiffness matrices for each element.

Element 1

For element 1, the angle between the global x and the local x' axes is 90° (counterclockwise) because x' is assumed to be directed from node 1 to node 2. Therefore,

$$C = \cos 90^\circ = \frac{x_2 - x_1}{L^{(1)}} = \frac{-60 - (-60)}{120} = 0$$

$$S = \sin 90^\circ = \frac{y_2 - y_1}{L^{(1)}} = \frac{120 - 0}{120} = 1$$

Also,

$$\begin{aligned} \frac{12I}{L^2} &= \frac{12(200)}{(10 \times 12)^2} = 0.167 \text{ in.}^2 \\ \frac{6I}{L} &= \frac{6(200)}{10 \times 12} = 10.0 \text{ in.}^3 \\ \frac{E}{L} &= \frac{30 \times 10^6}{10 \times 12} = 250,000 \text{ lb/in.}^3 \end{aligned} \quad (5.2.1)$$

Then, using Eqs. (5.2.1) to help in evaluating Eq. (5.1.11) for element 1, we obtain the element global stiffness matrix as

$$[k^{(1)}] = 250,000 \begin{bmatrix} u_1 & v_1 & \phi_1 & u_2 & v_2 & \phi_2 \\ 0.167 & 0 & -10 & -0.167 & 0 & -10 \\ 0 & 10 & 0 & 0 & -10 & 0 \\ -10 & 0 & 800 & 10 & 0 & 400 \\ -0.167 & 0 & 10 & 0.167 & 0 & 10 \\ 0 & -10 & 0 & 0 & 10 & 0 \\ -10 & 0 & 400 & 10 & 0 & 800 \end{bmatrix} \frac{\text{lb}}{\text{in.}} \quad (5.2.2)$$

where all diagonal terms are positive.

Element 2

For element 2, the angle between x and x' is zero because x' is directed from node 2 to node 3. Therefore,

$$C = 1 \quad S = 0$$

Also,

$$\begin{aligned}\frac{12I}{L^2} &= \frac{12(100)}{120^2} = 0.0835 \text{ in.}^2 \\ \frac{6I}{L} &= \frac{6(100)}{120} = 5.0 \text{ in.}^3 \\ \frac{E}{L} &= 250,000 \text{ lb/in.}^3\end{aligned}\quad (5.2.3)$$

Using the quantities obtained in Eqs. (5.2.3) in evaluating Eq. (5.1.11) for element 2, we obtain

$$[k^{(2)}] = 250,000 \begin{bmatrix} u_2 & v_2 & \phi_2 & u_3 & v_3 & \phi_3 \\ 10 & 0 & 0 & -10 & 0 & 0 \\ 0 & 0.0835 & 5 & 0 & -0.0835 & 5 \\ 0 & 5 & 400 & 0 & -5 & 200 \\ -10 & 0 & 0 & 10 & 0 & 0 \\ 0 & -0.0835 & -5 & 0 & 0.835 & -5 \\ 0 & 5 & 200 & 0 & -5 & 400 \end{bmatrix} \frac{\text{lb}}{\text{in.}} \quad (5.2.4)$$

Element 3

For element 3, the angle between x and x' is 270° (or -90°) because x' is directed from node 3 to node 4. Therefore,

$$C = 0 \quad S = -1$$

Therefore, evaluating Eq. (5.1.11) for element 3, we obtain

$$[k^{(3)}] = 250,000 \begin{bmatrix} u_3 & v_3 & \phi_3 & u_4 & v_4 & \phi_4 \\ 0.167 & 0 & 10 & -0.167 & 0 & 10 \\ 0 & 10 & 0 & 0 & -10 & 0 \\ 10 & 0 & 800 & -10 & 0 & 400 \\ -0.167 & 0 & -10 & 0.167 & 0 & -10 \\ -0 & -10 & 0 & 0 & 10 & 0 \\ 10 & 0 & 400 & -10 & 0 & 800 \end{bmatrix} \frac{\text{lb}}{\text{in.}} \quad (5.2.5)$$

Superposition of Eqs. (5.2.2), (5.2.4), and (5.2.5) and application of the boundary conditions $u_1 = v_1 = \phi_1 = 0$ and $u_4 = v_4 = \phi_4 = 0$ at nodes 1 and 4 yield the reduced set of equations for a longhand solution as

$$\begin{Bmatrix} 10,000 \\ 0 \\ 0 \\ 0 \\ 0 \\ 5000 \end{Bmatrix} = 250,000 \begin{bmatrix} 10.167 & 0 & 10 & -10 & 0 & 0 \\ 0 & 10.0835 & 5 & 0 & -0.0835 & 5 \\ 10 & 5 & 1200 & 0 & -5 & 200 \\ -10 & 0 & 0 & 10.167 & 0 & 10 \\ 0 & -0.0835 & -5 & 0 & 10.0835 & -5 \\ 0 & 5 & 200 & 10 & -5 & 1200 \end{bmatrix} \begin{Bmatrix} u_2 \\ v_2 \\ \phi_2 \\ u_3 \\ v_3 \\ \phi_3 \end{Bmatrix} \quad (5.2.6)$$

Solving Eq. (5.2.6) for the displacements and rotations, we have

$$\begin{Bmatrix} u_2 \\ v_2 \\ \phi_2 \\ u_3 \\ v_3 \\ \phi_3 \end{Bmatrix} = \begin{Bmatrix} 0.211 \text{ in.} \\ 0.00148 \text{ in.} \\ -0.00153 \text{ rad} \\ 0.209 \text{ in.} \\ -0.00148 \text{ in.} \\ -0.00149 \text{ rad} \end{Bmatrix} \quad (5.2.7)$$

The results indicate that the top of the frame moves to the right with negligible vertical displacement and small rotations of elements at nodes 2 and 3.

The element forces can now be obtained using $\{f'\} = [k'] [T] \{d\}$ for each element, as was previously done in solving truss and beam problems. We will illustrate this procedure only for element 1. For element 1, on using Eq. (5.1.10) for $[T]$ and Eq. (5.2.7) for the displacements at node 2, we have

$$[T]\{d\} = \begin{Bmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{Bmatrix} \begin{Bmatrix} u_1 = 0 \\ v_1 = 0 \\ \phi_1 = 0 \\ u_2 = 0.211 \\ v_2 = 0.00148 \\ \phi_2 = -0.00153 \end{Bmatrix} \quad (5.2.8)$$

On multiplying the matrices in Eq. (5.2.8), we obtain

$$[T]\{d\} = \begin{Bmatrix} 0 \\ 0 \\ 0 \\ 0.00148 \\ -0.211 \\ -0.00153 \end{Bmatrix} \quad (5.2.9)$$

Then using $[k']$ from Eq. (5.1.8), we obtain element 1 local forces as

$$\{f'\} = [k'] [T] \{d\} = 250,000 \begin{Bmatrix} 10 & 0 & 0 & -10 & 0 & 0 \\ 0 & 0.167 & 10 & 0 & -0.167 & 10 \\ 0 & 10 & 800 & 0 & -10 & 400 \\ -10 & 0 & 0 & 10 & 0 & 0 \\ 0 & -0.167 & -10 & 0 & 0.167 & -10 \\ 0 & 10 & 400 & 0 & -10 & 800 \end{Bmatrix} \begin{Bmatrix} 0 \\ 0 \\ 0 \\ 0.00148 \\ -0.211 \\ -0.00153 \end{Bmatrix} \quad (5.2.10)$$

Simplifying Eq. (5.2.10), we obtain the local forces acting on element 1 as

$$\begin{Bmatrix} f'_{1x} \\ f'_{1y} \\ m'_1 \\ f'_{2x} \\ f'_{2y} \\ m'_2 \end{Bmatrix} = \begin{Bmatrix} -3700 \text{ lb} \\ 4990 \text{ lb} \\ 376,000 \text{ lb-in.} \\ 3700 \text{ lb} \\ -4990 \text{ lb} \\ 223,000 \text{ lb-in.} \end{Bmatrix} \quad (5.2.11)$$

A free-body diagram of each element is shown in Figure 5–5 along with equilibrium verification. In Figure 5–5, the x' axis is directed from node 1 to node 2—consistent with the order of the nodal degrees of freedom used in developing the stiffness matrix for the element. Since the $x - y$ plane was initially established as shown in Figure 5–4, the z axis is directed outward—consequently, so is the z' axis (recall $z' = z$). The y' axis is then established such that x' cross y' yields the direction of z' . The signs on the resulting element forces in Eq. (5.2.11) are thus consistently shown in Figure 5–5. The forces in elements 2 and 3 can be obtained in a manner similar to that used to obtain Eq. (5.2.11) for the nodal forces in element 1. Here we report only the final results for the forces in elements 2 and 3 and leave it to your discretion to perform the detailed calculations. The element forces [shown in Figures 5–5(b) and (c)] are as follows:

Element 2

$$\begin{aligned} f'_{2x} &= 5010 \text{ lb} & f'_{2y} &= -3700 \text{ lb} & m'_2 &= -223,000 \text{ lb-in.} \\ f'_{3x} &= -5010 \text{ lb} & f'_{3y} &= 3700 \text{ lb} & m'_2 &= -221,000 \text{ lb-in.} \end{aligned} \quad (5.2.12a)$$

Element 3

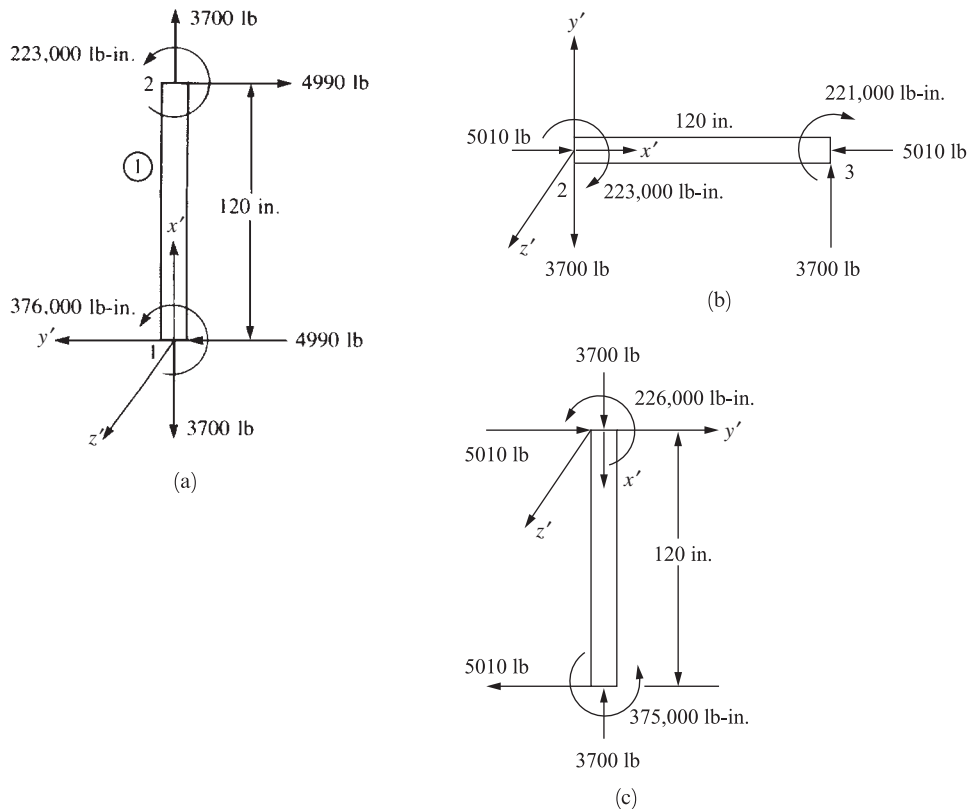
$$\begin{aligned} f'_{3x} &= 3700 \text{ lb} & f'_{3y} &= 5010 \text{ lb} & m'_3 &= 226,000 \text{ lb-in.} \\ f'_{4x} &= -3700 \text{ lb} & f'_{4y} &= -5010 \text{ lb} & m'_4 &= 375,000 \text{ lb-in.} \end{aligned} \quad (5.2.12b)$$

Considering the free body of element 1, the equilibrium equations are

$$\begin{aligned} \sum F_{x'} &: -4990 + 4990 = 0 \\ \sum F_{y'} &: -3700 + 3700 = 0 \\ \sum M_2 &: 376,000 + 223,000 - 4990(120 \text{ in.}) \cong 0 \end{aligned}$$

Considering moment equilibrium at node 2, we see from Eqs. (5.2.11) and (5.2.12a) that on element 1, $m'_2 = 223,000 \text{ lb-in.}$, and the opposite value, $-223,000 \text{ lb-in.}$, occurs on element 2. Similarly, moment equilibrium is satisfied at node 3, as m'_3 from elements 2 and 3 add to the 5000 lb-in. applied moment. That is, from Eqs. (5.2.12a) and (5.2.12b) we have

$$-221,000 + 226,000 = 5000 \text{ lb-in.}$$



■ **Figure 5-5** Free-body diagrams of (a) element 1, (b) element 2, and (c) element 3

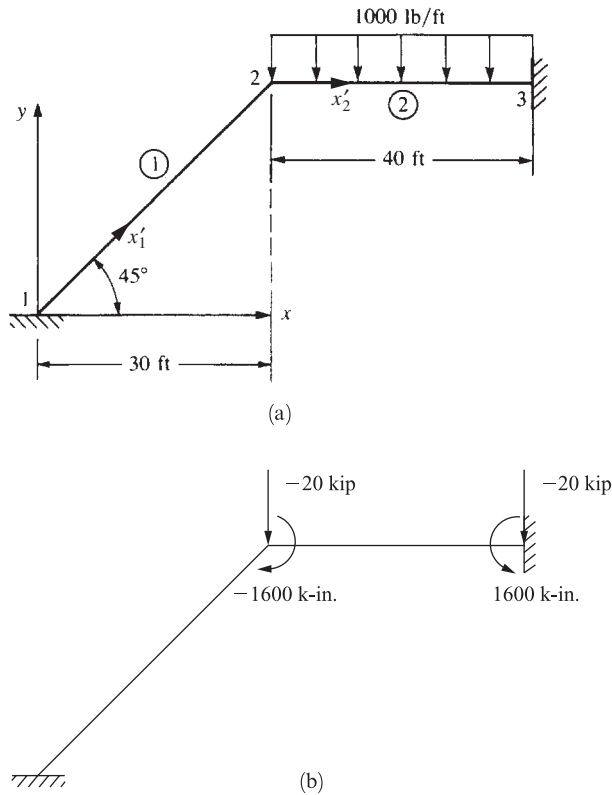
EXAMPLE 5.2

To illustrate the procedure for solving frames subjected to distributed loads, solve the rigid plane frame shown in Figure 5-6. The frame is fixed at nodes 1 and 3 and subjected to a uniformly distributed load of 1000 lb/ft applied downward over element 2. The global-coordinate axes have been established at node 1. The element lengths are shown in the figure. Let $E = 30 \times 10^6$ psi, $A = 100$ in.² and $I = 1000$ in.⁴ for both elements of the frame.

SOLUTION:

We begin by replacing the distributed load acting on element 2 by nodal forces and moments acting at nodes 2 and 3. Using Eqs. (4.4.5)–(4.4.7) (or Appendix D), the equivalent nodal forces and moments are calculated as

$$\begin{aligned}
 f_{2y} &= -\frac{wL}{2} = -\frac{(1000)40}{2} = -20,000 \text{ lb} = -20 \text{ kip} \\
 m_2 &= -\frac{wL^2}{12} = -\frac{(1000)40^2}{12} = -133,333 \text{ lb-ft} = -1600 \text{ k-in.}
 \end{aligned}
 \tag{5.2.13}$$



■ **Figure 5-6** (a) Plane frame for analysis and (b) equivalent nodal forces on frame

$$f_{3y} = -\frac{wL}{2} = -\frac{(1000)40}{2} = -20,000 \text{ lb} = -20 \text{ kip}$$

$$m_3 = \frac{wL^2}{12} = \frac{(1000)40^2}{12} = 133,333 \text{ lb-ft} = 1600 \text{ k-in.}$$

We then use Eq. (5.1.11) to determine each element stiffness matrix:

Element 1

$$\theta^{(1)} = 45^\circ \quad C = 0.707 \quad S = 0.707 \quad L^{(1)} = 42.4 \text{ ft} = 509.0 \text{ in.}$$

$$\frac{E}{L} = \frac{30 \times 10^3}{509} = 58.93$$

$$[k^{(1)}] = 58.93 \begin{bmatrix} 50.02 & 49.98 & 8.33 \\ 49.98 & 50.02 & -8.33 \\ 8.33 & -8.33 & 4000 \end{bmatrix} \frac{\text{kip}}{\text{in.}} \quad (5.2.14)$$

Simplifying Eq. (5.2.14), we obtain

$$[k^{(1)}] = \begin{bmatrix} u_2 & v_2 & \phi_2 \\ 2948 & 2945 & 491 \\ 2945 & 2948 & -491 \\ 491 & -491 & 235,700 \end{bmatrix} \frac{\text{kip}}{\text{in.}} \quad (5.2.15)$$

where only the lower right 3×3 submatrix of Eq. (5.1.11) of the stiffness matrix associated with degrees of freedom at node 2 are included because node 1 is fixed.

Element 2

$$\theta^{(2)} = 0^\circ \quad C = 1 \quad S = 0 \quad L^{(2)} = 40 \text{ ft} = 480 \text{ in.}$$

$$\frac{E}{L} = \frac{30 \times 10^3}{480} = 62.50$$

$$[k^{(2)}] = 62.50 \begin{bmatrix} 100 & 0 & 0 \\ 0 & 0.052 & 12.5 \\ 0 & 12.5 & 4000 \end{bmatrix} \frac{\text{kip}}{\text{in.}} \quad (5.2.16)$$

Simplifying Eq. (5.2.16), we obtain

$$[k^{(2)}] = \begin{bmatrix} u_2 & v_2 & \phi_2 \\ 6250 & 0 & 0 \\ 0 & 3.25 & 781.25 \\ 0 & 781.25 & 250,00 \end{bmatrix} \frac{\text{kip}}{\text{in.}} \quad (5.2.17)$$

where, only the upper left 3×3 submatrix of Eq. (5.1.11) of the stiffness matrix associated with degrees of freedom at node 2 are included because node 3 is fixed. On superimposing the stiffness matrices of the elements, using Eqs. (5.2.15) and (5.2.17), and using Eq. (5.2.13) for the nodal forces and moments only at node 2 (because the structure is fixed at node 3), we have

$$\begin{Bmatrix} F_{2x} = 0 \\ F_{2y} = -20 \\ M_2 = -1600 \end{Bmatrix} = \begin{bmatrix} 9198 & 2945 & 491 \\ 2945 & 2951 & 290 \\ 491 & 290 & 485,700 \end{bmatrix} \begin{Bmatrix} u_2 \\ v_2 \\ \phi_2 \end{Bmatrix} \quad (5.2.18)$$

Solving Eq. (5.2.18) for the displacements and the rotation at node 2, we obtain

$$\begin{Bmatrix} u_2 \\ v_2 \\ \phi_2 \end{Bmatrix} = \begin{Bmatrix} 0.0033 \text{ in.} \\ -0.0097 \text{ in.} \\ -0.0033 \text{ rad} \end{Bmatrix} \quad (5.2.19)$$

The results indicate that node 2 moves to the right ($u_2 = 0.0033 \text{ in.}$) and down ($v_2 = -0.0097 \text{ in.}$) and the rotation of the joint is clockwise ($\phi_2 = -0.0033 \text{ rad}$).

The local forces in each element can now be determined. The procedure for elements that are subjected to a distributed load must be applied to element 2. Recall that the local forces are given by $\{f'\} = [k'][T]\{d\}$. For element 1, we then have

$$[T]\{d\} = \begin{bmatrix} 0.707 & 0.707 & 0 & 0 & 0 & 0 \\ -0.707 & 0.707 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0.707 & 0.707 & 0 \\ 0 & 0 & 0 & -0.707 & 0.707 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{Bmatrix} 0 \\ 0 \\ 0 \\ 0.0033 \\ -0.0097 \\ -0.0033 \end{Bmatrix} \quad (5.2.20)$$

Simplifying Eq. (5.2.20) yields

$$[T]\{d\} = \begin{Bmatrix} 0 \\ 0 \\ 0 \\ -0.00452 \\ -0.0092 \\ -0.0033 \end{Bmatrix} \quad (5.2.21)$$

Using Eq. (5.2.21) and Eq. (5.1.8) for $[k']$, we obtain

$$\begin{Bmatrix} f'_{1x} \\ f'_{1y} \\ m'_1 \\ f'_{2x} \\ f'_{2y} \\ m'_2 \end{Bmatrix} = \begin{bmatrix} 5893 & 0 & 0 & -5893 & 0 & 0 \\ & 2.730 & 694.8 & 0 & -2.730 & 694.8 \\ & & 235,800 & 0 & -694.8 & 117,900 \\ & & & 5893 & 0 & 0 \\ & & & & 2.730 & -694.8 \\ \text{Symmetry} & & & & & 235,800 \end{bmatrix} \begin{Bmatrix} 0 \\ 0 \\ 0 \\ -0.00452 \\ -0.0092 \\ -0.0033 \end{Bmatrix} \quad (5.2.22)$$

Simplifying Eq. (5.2.22) yields the local forces in element 1 as

$$\begin{aligned} f'_{1x} &= 26.64 \text{ kip} & f'_{1y} &= -2.268 \text{ kip} & m'_1 &= -389.1 \text{ k-in.} \\ f'_{2x} &= -26.64 \text{ kip} & f'_{2y} &= 2.268 \text{ kip} & m'_2 &= -778.2 \text{ k-in.} \end{aligned} \quad (5.2.23)$$

For element 2, the local forces are given by Eq. (4.4.11) because a distributed load is acting on the element. From Eqs. (5.1.10) and (5.2.19), we then have

$$[T]\{d\} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{Bmatrix} 0.0033 \\ -0.0097 \\ -0.0033 \\ 0 \\ 0 \\ 0 \end{Bmatrix} \quad (5.2.24)$$

Simplifying Eq. (5.2.24), we obtain

$$[T]\{d\} = \begin{Bmatrix} 0.0033 \\ -0.0097 \\ -0.0033 \\ 0 \\ 0 \\ 0 \end{Bmatrix} \quad (5.2.25)$$

Using Eq. (5.2.25) and Eq. (5.1.8) for $[k']$, we have

$$[k']\{d'\} = [k'] [T]\{d\} = \begin{bmatrix} 6250 & 0 & 0 & -6250 & 0 & 0 \\ & 3.25 & 781.1 & 0 & -3.25 & 781.1 \\ & & 250,000 & 0 & -781.1 & 125,000 \\ & & & 6250 & 0 & 0 \\ & & & & 3.25 & -781.1 \\ \text{Symmetry} & & & & & 250,000 \end{bmatrix} \begin{Bmatrix} 0.0033 \\ -0.0097 \\ -0.0033 \\ 0 \\ 0 \\ 0 \end{Bmatrix} \quad (5.2.26)$$

Simplifying Eq. (5.2.26) yields

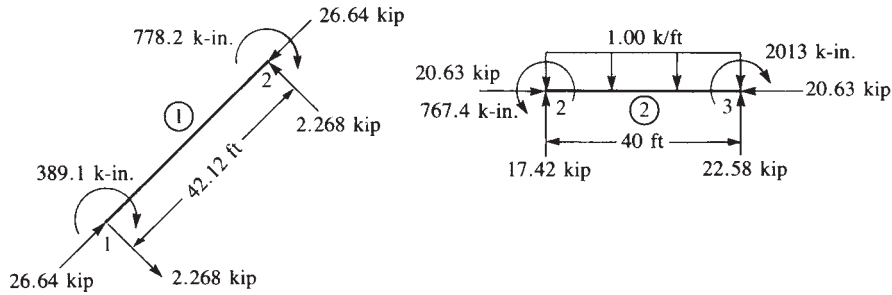
$$[k']\{d'\} = \begin{Bmatrix} 20.63 \\ -2.58 \\ -832.57 \\ -20.63 \\ 2.58 \\ -412.50 \end{Bmatrix} \quad (5.2.27)$$

To obtain the actual element local nodal forces, we apply Eq. (4.4.11); that is, we must subtract the equivalent nodal forces [Eqs. (5.2.13)] from Eq. (5.2.27) to yield

$$\begin{Bmatrix} f'_{2x} \\ f'_{2y} \\ m'_2 \\ f'_{3x} \\ f'_{3y} \\ m'_3 \end{Bmatrix} = \begin{Bmatrix} 20.63 \\ -2.58 \\ -832.57 \\ -20.63 \\ 2.58 \\ -412.50 \end{Bmatrix} - \begin{Bmatrix} 0 \\ -20 \\ -1600 \\ 0 \\ -20 \\ 1600 \end{Bmatrix} \quad (5.2.28)$$

Simplifying Eq. (5.2.28), we obtain

$$\begin{aligned} f'_{2x} &= 20.63 \text{ kip} & f'_{2y} &= 17.42 \text{ kip} & m'_2 &= 767.4 \text{ k-in.} \\ f'_{3x} &= -20.63 \text{ kip} & f'_{3y} &= 22.58 \text{ kip} & m'_3 &= -2013 \text{ k-in.} \end{aligned} \quad (5.2.29)$$



■ **Figure 5-7** Free-body diagrams of elements 1 and 2

Using Eqs. (5.2.23) and (5.2.29) for the local forces in each element, we can construct the free-body diagram for each element, as shown in Figure 5-7. From the free-body diagrams, one can confirm the equilibrium of each element, the total frame, and joint 2 as desired.

In Example 5.3, we will illustrate the equivalent joint force replacement method for a frame subjected to a load acting on an element instead of at one of the joints of the structure. Since no distributed loads are present, the point of application of the concentrated load could be treated as an extra joint in the analysis, and we could solve the problem in the same manner as Example 5.1.

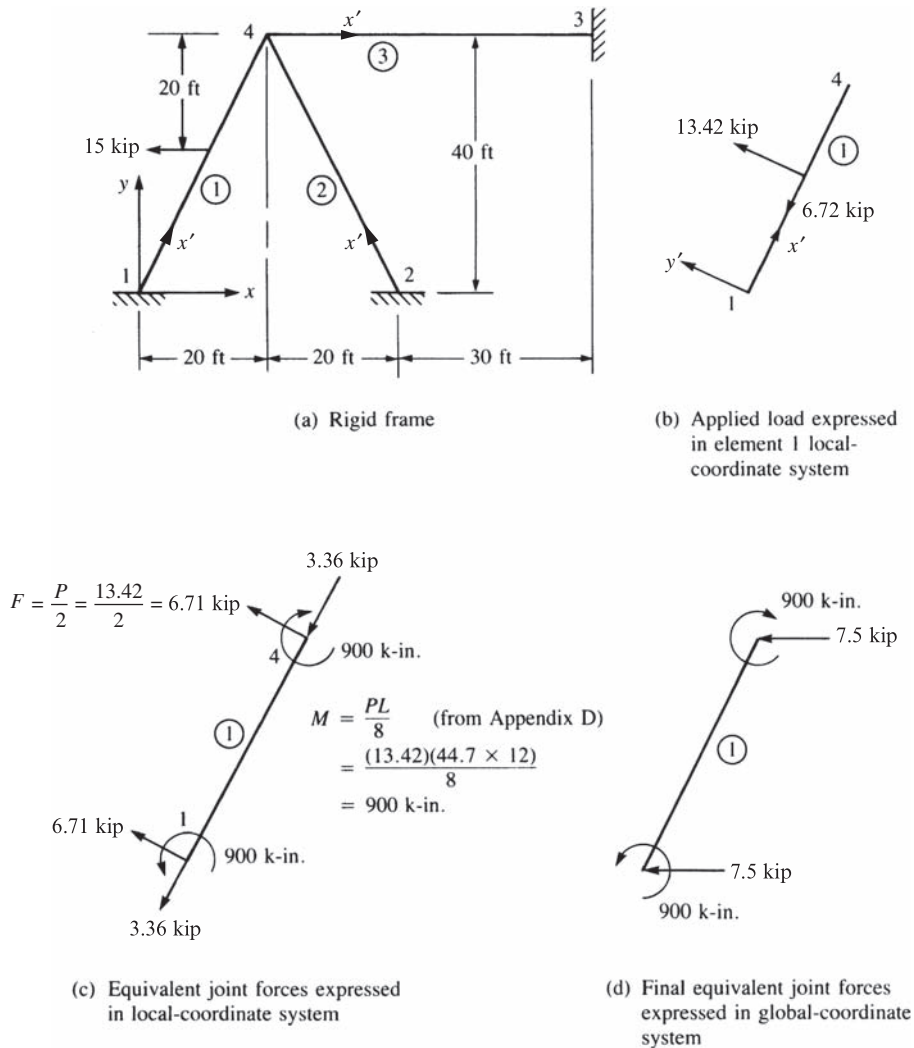
This approach has the disadvantage of increasing the total number of joints, as well as the size of the total structure stiffness matrix $[K]$. For small structures solved by computer, this does not pose a problem. However, for very large structures, this might reduce the maximum size of the structure that could be analyzed. Certainly, this additional node greatly increases the longhand solution time for the structure. Hence, we will illustrate a standard procedure based on the concept of equivalent joint forces applied to the case of concentrated loads. We will again use Appendix D.

EXAMPLE 5.3

Solve the frame shown in Figure 5-8(a). The frame consists of the three elements shown and is subjected to a 15-kip horizontal load applied at midlength of element 1. Nodes 1, 2, and 3 are fixed, and the dimensions are shown in the figure. Let $E = 30 \times 10^6$ psi, $I = 800$ in.⁴, and $A = 8$ in.² for all elements.

SOLUTION:

1. We first express the applied load in the element 1 local coordinate system (here x' is directed from node 1 to node 4). This is shown in Figure 5-8(b).
2. Next, we determine the equivalent joint forces $\{f_0\}$ at each end of element 1, using the table in Appendix D. (These forces are of opposite sign from what are traditionally known as *fixed-end forces* in classical structural analysis theory [1].) These equivalent forces (and moments) are shown in Figure 5-8(c).



■ **Figure 5-8** Rigid frame with a load applied on an element

3. We then transform the equivalent joint forces from the present local-coordinate-system forces into the global-coordinate-system forces, using the equation $\{f\} = [T]^T \{f'\}$, where $[T]$ is defined by Eq. (5.1.10). These global joint forces are shown in Figure 5-8(d).
4. Then we analyze the structure in Figure 5-8(d), using the equivalent joint forces (plus actual joint forces, if any) in the usual manner.
5. We obtain the final internal forces developed at the ends of each element that has an applied load (here element 1 only) by subtracting step 2 joint forces from step 4 joint forces; that is, Eq. (4.4.11) ($\{f\} = \{f^{(e)}\} - \{f_0\}$) is applied locally to all elements that originally had loads acting on them.

The solution of the structure as shown in Figure 5-8(d) now follows. Using Eq. (5.1.11), we obtain the global stiffness matrix for each element.

Element 1

For element 1, the angle between the global x and the local x' axes is 63.43° because x' is assumed to be directed from node 1 to node 4. Therefore,

$$\begin{aligned}
 C &= \cos 63.43^\circ = \frac{x_4 - x_1}{L^{(1)}} = \frac{20 - 0}{44.7} = 0.447 \\
 S &= \sin 63.43^\circ = \frac{y_4 - y_1}{L^{(1)}} = \frac{40 - 0}{44.7} = 0.895 \\
 \frac{12I}{L^2} &= \frac{12(800)}{(44.7 \times 12)^2} = 0.334 \quad \frac{6I}{L} = \frac{6(800)}{44.7 \times 12} = 8.95 \\
 \frac{E}{L} &= \frac{30 \times 10^3}{44.7 \times 12} = 55.9
 \end{aligned}$$

Using the preceding results in Eq. (5.1.11) for $[k]$, we obtain

$$[k^{(1)}] = \begin{bmatrix} u_4 & v_4 & \phi_4 \\ 90.9 & 178 & 448 \\ 178 & 359 & -224 \\ 448 & -224 & 179,000 \end{bmatrix} \quad (5.2.30)$$

where only the lower right 3×3 submatrix of Eq. (5.1.11) of the stiffness matrix associated with degrees of freedom at node 4 are included because node 1 is fixed and, hence, not needed in the solution for the nodal displacements.

Element 3

For element 3, the angle between x and x' is zero because x' is assumed to be directed from node 4 to node 3. Therefore,

$$\begin{aligned}
 C &= 1 \quad S = 0 \quad \frac{12I}{L^2} = \frac{12(800)}{(50 \times 12)^2} = 0.0267 \\
 \frac{6I}{L} &= \frac{6(800)}{50 \times 12} = 8.00 \quad \frac{E}{L} = \frac{30 \times 10^3}{50 \times 12} = 50
 \end{aligned}$$

Substituting these results into the upper left 3×3 submatrix of Eq. (5.1.11) of $[k]$, we obtain

$$[k^{(3)}] = \begin{bmatrix} u_4 & v_4 & \phi_4 \\ 400 & 0 & 0 \\ 0 & 1.334 & 400 \\ 0 & 400 & 160,000 \end{bmatrix} \quad (5.2.31)$$

since node 3 is fixed.

Element 2

For element 2, the angle between x and x' is 116.57° because x' is assumed to be directed from node 2 to node 4. Therefore,

$$C = \frac{20 - 40}{44.7} = -0.447 \quad S = \frac{40 - 0}{44.7} = 0.895$$

$$\frac{12I}{L^2} = 0.0334 \quad \frac{6I}{L} = 8.95 \quad \frac{E}{L} = 55.9$$

since element 2 has the same properties as element 1. Substituting these results into the lower right 3×3 submatrix of Eq. (5.1.11) of $[k]$, we obtain

$$[k^{(2)}] = \begin{bmatrix} & u_4 & v_4 & \phi_4 \\ 90.9 & -178 & 448 \\ -178 & 359 & 224 \\ 448 & 224 & 179,000 \end{bmatrix} \quad (5.2.32)$$

since node 2 is fixed. On superimposing the stiffness matrices given by Eqs. (5.2.30), (5.2.31), and (5.2.32), and using the nodal forces given in Figure 5–8(d) at node 4 only, we have

$$\begin{Bmatrix} -7.50 \text{ kip} \\ 0 \\ -900 \text{ k-in.} \end{Bmatrix} = \begin{bmatrix} 582 & 0 & 896 \\ 0 & 719 & 400 \\ 896 & 400 & 518,000 \end{bmatrix} \begin{Bmatrix} u_4 \\ v_4 \\ \phi_4 \end{Bmatrix} \quad (5.2.33)$$

Simultaneously solving the three equations in Eq. (5.2.33), we obtain

$$\begin{aligned} u_4 &= -0.0103 \text{ in.} \\ v_4 &= 0.000956 \text{ in.} \\ \phi_4 &= -0.00172 \text{ rad} \end{aligned} \quad (5.2.34)$$

Next, we determine the element forces by again using $\{f'\} = [k'][T]\{d\}$. In general, we have

$$[T]\{d\} = \begin{bmatrix} C & S & 0 & 0 & 0 & 0 \\ -S & C & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & C & S & 0 \\ 0 & 0 & 0 & -S & C & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{Bmatrix} u_i \\ v_i \\ \phi_i \\ u_j \\ v_j \\ \phi_j \end{Bmatrix}$$

Thus, the preceding matrix multiplication yields

$$[T]\{d\} = \begin{Bmatrix} Cu_i + Sv_i \\ -Su_i + Cv_i \\ \phi_i \\ Cu_j + Sv_j \\ -Su_j + Cv_j \\ \phi_j \end{Bmatrix} \quad (5.2.35)$$

Element 1

$$[T]\{d\} = \begin{Bmatrix} 0 \\ 0 \\ 0 \\ (0.447)(-0.0103) + (0.895)(0.000956) \\ (-0.895)(-0.0103) + (0.447)(0.000956) \\ -0.00172 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \\ 0 \\ -0.00374 \\ 0.00963 \\ -0.00172 \end{Bmatrix} \quad (5.2.36)$$

Using Eq. (5.1.8) for $[k']$ and Eq. (5.2.36), we obtain

$$[k'] [T]\{d\} = \begin{bmatrix} 447 & 0 & 0 & -447 & 0 & 0 \\ 0 & 1.868 & 500.5 & 0 & -1.868 & 500.5 \\ 0 & 500.5 & 179,000 & 0 & -500.5 & 89,490 \\ -447 & 0 & 0 & 447 & 0 & 0 \\ 0 & -1.868 & -500.5 & 0 & 1.868 & -500.5 \\ 0 & 500.5 & 89,490 & 0 & -500.5 & 179,000 \end{bmatrix} \times \begin{Bmatrix} 0 \\ 0 \\ 0 \\ -0.00374 \\ 0.00963 \\ -0.00172 \end{Bmatrix} \quad (5.2.37)$$

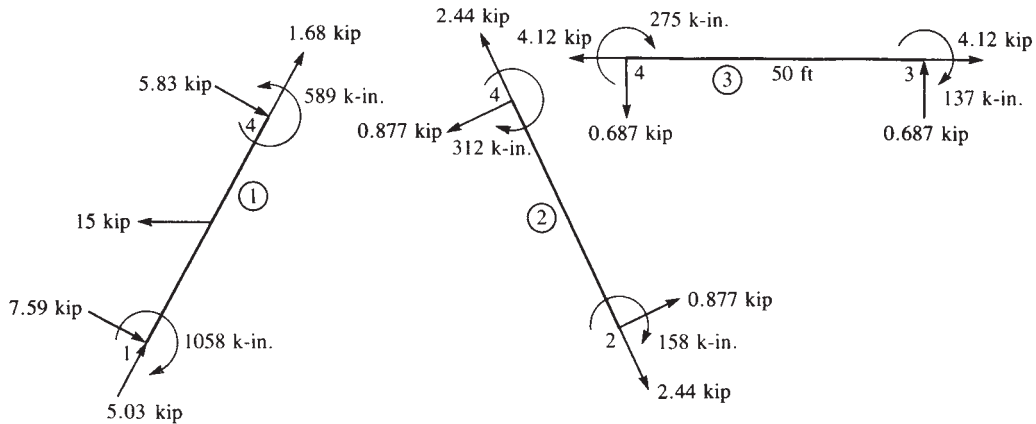
These values are now called *effective nodal forces* $\{f^{(e)}\}$. Multiplying the matrices of Eq. (5.2.37) and using Eq. (4.4.11) to subtract the equivalent nodal forces in local coordinates for the element shown in Figure 5–8(c), we obtain the final nodal forces in element 1 as

$$\{f^{(1)}\} = \begin{Bmatrix} 1.67 \\ -0.88 \\ -158 \\ -1.67 \\ 0.88 \\ -311 \end{Bmatrix} - \begin{Bmatrix} -3.36 \\ 6.71 \\ 900 \\ -3.36 \\ 6.71 \\ -900 \end{Bmatrix} = \begin{Bmatrix} 5.03 \text{ kip} \\ -7.59 \text{ kip} \\ -1058 \text{ k-in.} \\ 1.68 \text{ kip} \\ -5.83 \text{ kip} \\ 589 \text{ k-in.} \end{Bmatrix} \quad (5.2.38)$$

Similarly, we can use Eqs. (5.2.35) and (5.1.8) for elements 3 and 2 to obtain the local nodal forces in these elements. Since these elements do not have any applied loads on them, the final nodal forces in local coordinates associated with each element are given by $\{f'\} = [k'] [T]\{d\}$. These forces have been determined as follows:

Element 3

$$\begin{aligned} f'_{4x} &= -4.12 \text{ kip} & f'_{4y} &= -0.687 \text{ kip} & m'_4 &= -275 \text{ k-in.} \\ f'_{3x} &= 4.12 \text{ kip} & f'_{3y} &= 0.687 \text{ kip} & m'_3 &= -137 \text{ k-in.} \end{aligned} \quad (5.2.39)$$



■ **Figure 5-9** Free-body diagrams of all elements of the frame in Figure 5-8(a)

Element 2

$$\begin{aligned} f'_{2x} &= -2.44 \text{ kip} & f'_{2y} &= -0.877 \text{ kip} & m'_2 &= -158 \text{ k-in.} \\ f'_{4x} &= 2.44 \text{ kip} & f'_{4y} &= 0.877 \text{ kip} & m'_4 &= -312 \text{ k-in.} \end{aligned} \quad (5.2.40)$$

Free-body diagrams of all elements are shown in Figure 5-9. Each element has been determined to be in equilibrium, as often occurs even if errors are made in the longhand calculations. However, equilibrium at node 4 and equilibrium of the whole frame are also satisfied. For instance, using the results of Eqs. (5.2.38) through (5.2.40) to check equilibrium at node 4, which is implicit in the formulation of the global equations, we have

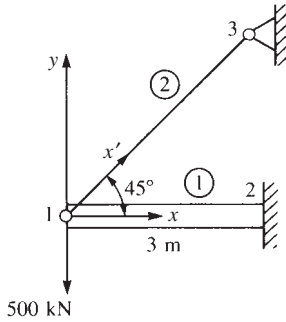
$$\begin{aligned} \sum M_4 &= 589 - 275 - 312 = 2 \text{ k-in.} && \text{(close to zero)} \\ \sum F_x &= 1.68(0.447) + 5.83(0.895) - 2.44(0.447) \\ &\quad - 0.877(0.895) - 4.12 = -0.027 \text{ kip} && \text{(close to zero)} \\ \sum F_y &= 1.68(0.895) - 5.83(0.447) + 2.44(0.895) \\ &\quad - 0.877(0.447) - 0.687 = 0.004 \text{ kip} && \text{(close to zero)} \end{aligned}$$

Thus, the solution has been verified to be correct within the accuracy associated with a longhand solution.

To illustrate the solution of a problem involving both bar and frame elements, we will solve the following example.

EXAMPLE 5.4

The bar element 2 is used to stiffen the cantilever beam element 1, as shown in Figure 5-10. Determine the displacements at node 1 and the element forces. For the bar, let $A = 1.0 \times 10^{-3} \text{ m}^2$. For the beam, let $A = 2 \times 10^{-3} \text{ m}^2$, $I = 5 \times 10^{-5} \text{ m}^4$, and $L = 3 \text{ m}$. For both the bar and the beam elements, let $E = 210 \text{ GPa}$. Let the angle between the beam and the bar be 45° . A downward force of 500 kN is applied at node 1.



■ **Figure 5–10** Cantilever beam with a bar element support

SOLUTION:

For brevity's sake, since nodes 2 and 3 are fixed, we keep only the parts of $[k]$ for each element that are needed to obtain the global $[K]$ matrix necessary for solution of the nodal degrees of freedom. Using Eq. (3.4.23), we obtain $[k]$ for the bar as

$$[k^{(2)}] = \frac{(1 \times 10^{-3})(210 \times 10^6)}{(3/\cos 45^\circ)} \begin{bmatrix} 0.5 & 0.5 \\ 0.5 & 0.5 \end{bmatrix}$$

or, simplifying this equation, we obtain

$$[k^{(2)}] = 70 \times 10^3 \begin{bmatrix} u_1 & v_1 \\ 0.354 & 0.354 \\ 0.354 & 0.354 \end{bmatrix} \frac{\text{kN}}{\text{m}} \quad (5.2.41)$$

Using the upper left 3×3 part of Eq. (5.1.11), we obtain $[k]$ for the beam (including axial effects) as

$$[k^{(1)}] = 70 \times 10^3 \begin{bmatrix} u_1 & v_1 & \phi_1 \\ 2 & 0 & 0 \\ 0 & 0.067 & 0.10 \\ 0 & 0.10 & 0.20 \end{bmatrix} \frac{\text{kN}}{\text{m}} \quad (5.2.42)$$

where $(E/L) \times 10^{-3}$ has been factored out in evaluating Eq. (5.2.42).

We assemble Eqs. (5.2.41) and (5.2.42) in the usual manner to obtain the global stiffness matrix as

$$[K] = 70 \times 10^3 \begin{bmatrix} 2.354 & 0.354 & 0 \\ 0.354 & 0.421 & 0.10 \\ 0 & 0.10 & 0.20 \end{bmatrix} \frac{\text{kN}}{\text{m}} \quad (5.2.43)$$

The global equations are then written for node 1 as

$$\begin{Bmatrix} F_{1x} \\ F_{1y} \\ M_1 \end{Bmatrix} = \begin{Bmatrix} 0 \\ -500 \\ 0 \end{Bmatrix} = 70 \times 10^3 \begin{bmatrix} 2.354 & 0.354 & 0 \\ 0.354 & 0.421 & 0.10 \\ 0 & 0.10 & 0.20 \end{bmatrix} \begin{Bmatrix} u_1 \\ v_1 \\ \phi_1 \end{Bmatrix} \quad (5.2.44)$$

Solving Eq. (5.2.44), we obtain

$$u_1 = 0.00338 \text{ m} \quad v_1 = -0.0225 \text{ m} \quad \phi_1 = 0.0113 \text{ rad} \quad (5.2.45)$$

In general, the local element forces are obtained using $\{f'\} = [k'][T]\{d\}$. For the bar element, we then have

$$\begin{Bmatrix} f'_{1x} \\ f'_{3x} \end{Bmatrix} = \frac{AE}{L} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} C & S & 0 & 0 \\ 0 & 0 & C & S \end{bmatrix} \begin{Bmatrix} u_1 \\ v_1 \\ u_3 \\ v_3 \end{Bmatrix} \quad (5.2.46)$$

The matrix triple product of Eq. (5.2.46) yields (as one equation)

$$f'_{1x} = \frac{AE}{L}(Cu_1 + Sv_1) \quad (5.2.47)$$

Substituting the numerical values into Eq. (5.2.47), we obtain

$$f'_{1x} = \frac{(1 \times 10^{-3} \text{ m}^2)(210 \times 10^6 \text{ kN/m}^2)}{4.24 \text{ m}} \left[\frac{\sqrt{2}}{2}(0.00338 - 0.0225) \right] \quad (5.2.48)$$

Simplifying Eq. (5.2.48), we obtain the axial force in the bar (element 2) as

$$f'_{1x} = -670 \text{ kN} \quad (5.2.49)$$

where the negative sign means f'_{1x} is in the direction opposite x' for element 2. Similarly, we obtain

$$f'_{3x} = 670 \text{ kN} \quad (5.2.50)$$

which means the bar is in tension as shown in Figure 5–11. Since the local and global axes are coincident for the beam element, we have $\{f'\} = \{f\}$ and $\{d'\} = \{d\}$. Therefore, from Eq. (5.1.6), we have at node 1

$$\begin{Bmatrix} f'_{1x} \\ f'_{1y} \\ m'_1 \end{Bmatrix} = \begin{bmatrix} C_1 & 0 & 0 \\ 0 & 12C_2 & 6C_2L \\ 0 & 6C_2L & 4C_2L^2 \end{bmatrix} \begin{Bmatrix} u_1 \\ v_1 \\ \phi_1 \end{Bmatrix} \quad (5.2.51)$$

where only the upper left 3×3 part of the stiffness matrix is needed because the displacements at node 2 are equal to zero. Substituting numerical values into Eq. (5.2.51), we obtain

$$\begin{Bmatrix} f'_{1x} \\ f'_{1y} \\ m'_1 \end{Bmatrix} = 70 \times 10^3 \begin{bmatrix} 2 & 0 & 0 \\ 0 & 0.067 & 0.10 \\ 0 & 0.10 & 0.20 \end{bmatrix} \begin{Bmatrix} 0.00338 \\ -0.0225 \\ 0.0113 \end{Bmatrix}$$

The matrix product then yields

$$f'_{1x} = 473 \text{ kN} \quad f'_{1y} = -26.5 \text{ kN} \quad m'_1 = 0.0 \text{ kN} \cdot \text{m} \quad (5.2.52)$$